

Problem

A certain network has an input admittance $Y(s)$. The admittance has a pole at $s = -3$, a zero at $s = -1$, and $Y(\infty) = 0.25 \text{ S}$.

(a) Find $Y(s)$.

(b) An 8-V battery is connected to the network via a switch. If the switch is closed at $t = 0$, find the current $i(t)$ through $Y(s)$ using the Laplace transform.

Step-by-step solution

Step 1 of 3

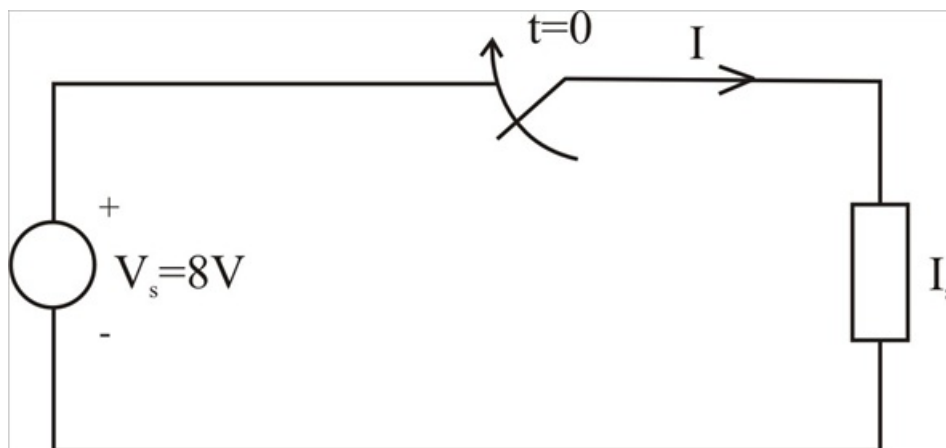
(a) Let $Y(s) = \frac{K(s+1)}{s+3}$

$$Y(\infty) \lim_{s \rightarrow \infty} \frac{K(s+1)}{s+3} = \lim_{s \rightarrow \infty} \frac{K\left(1 + \frac{1}{s}\right)}{s + \frac{3}{s}} = K$$

Hence $Y(s) = \frac{s+1}{4(s+3)}$

Step 2 of 3

(b) Consider the circuit shown below



Step 3 of 3

$$V_s = 8u(t) \rightarrow V_s = \frac{8}{s}$$

$$I = \frac{V_s}{Z} = Y(s)V_s(s) = \frac{8}{4s} \cdot \frac{s+1}{s+3} = \frac{2(s+1)}{s(s+3)}$$

$$I = \frac{A}{s} + \frac{B}{s+3}$$

$$A = \frac{2}{3} \qquad B = \frac{-4}{3}$$

$$i(t) = \frac{1}{3} [2 - 4e^{-3t}] u(t) \quad A$$